Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	28-10-2026	**Team ID**	**EN25CS3011021**
Project Name	EV Market Performance and Insights	**Maximum Marks**	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	Connect Tableau to the EV market dataset (CSV / Excel) covering vehicle sales, price, battery specifications, driving range, manufacturer, manufacturing country, customer ratings, market region and CO2 metrics; relate or join the tables on common fields.	High
2	Data Processing	Clean and standardize the data (data types, null values, duplicates, category names); assign geographic roles to country and region; prepare fields such as battery type, drivetrain, vehicle category and model year for analysis.	High
3	Business Analysis	Calculate the KPIs and measures: total EV sales, average vehicle price, average vehicle range, CO2 saved (tons per year), 3-year retained value, customer rating, battery cycle life, maximum DC charging power and energy efficiency (Wh per km); aggregate them by category, manufacturer, battery type, country, region and year.	High
4	Business Insights & Analytical Outputs	Provide two interactive dashboards. Dashboard 1: KPI tiles, production output by country and category, top manufacturers by customer review and sales units, driving range distribution, and 3-year retained value. Dashboard 2: sales trend over time, battery technology performance, charging vs efficiency, and regional CO2 saved by drivetrain.	High
5	User Interaction & Analysis	Let users filter by vehicle category through the shared legend, highlight and sort data, use tooltips, and compare categories, manufacturers, battery technologies, countries and regions across the charts.	Medium

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
6	Data Update & Reporting	Refresh the data source so that all worksheets and dashboards update, and make the dashboards shareable and exportable (image / PDF) for executive and stakeholder reporting.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	Dashboards use clear titles, a consistent color legend by vehicle category, tooltips and interactive filters so executives and analysts can understand insights without training.
NFR-2	Security	Access to the dataset and workbook is limited to authorized users, and dashboards are shared with view-only permission. The data covers market-level information and contains no personal customer data.
NFR-3	Reliability	Dashboards show accurate and consistent values after every refresh; calculated fields are validated against the source data and data connection errors are reported clearly.
NFR-4	Performance	Dashboards load and respond to filters within a few seconds by using extracts, aggregated measures and a limited number of marks per chart.
NFR-5	Availability	Published dashboards are accessible online to stakeholders whenever needed, and a packaged workbook copy is kept as a backup.
NFR-6	Scalability	The design supports more rows, new model years, manufacturers and vehicle categories by refreshing the data extract, without redesigning the dashboards.